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MULTIOMICS DATA SCIENCE - (TIME) SERIES ANALYSIS

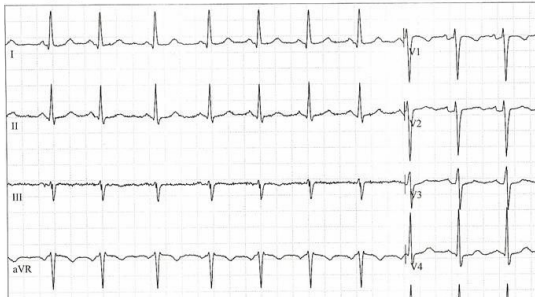
(TIME)SERIES

- A series is a (sorted) collection of dependent numerical values
- A time series maps time points (x axis) on another (physical) dependent value with arbitrary unit

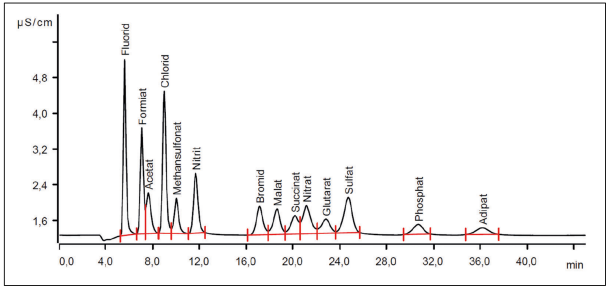
(TIME)SERIES

- A series is a (sorted) collection of dependent numerical values
- A time series maps time points (x axis) on another (physical) dependent value with arbitrary unit
- Examples:
  - Stock price
  - Electrocardiogram (EKG)
  - Chromatograms

Quelle: https://www.finanzen.at/chart/tesla



Quelle: https://de.wikipedia.org/wiki/Elektrokardiogramm



Quelle: https://www.all-electronics.de

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OTHER SERIES

- Numerical series, examples:
  - Weight
  - Length
  - Frequences

- Set with independent values:
  - Body heights of 1000 people
  - Concentrations of 4000 identified proteins

- Nominal / categorial values:
  - Weight: below average, average, above average, obese
  - Times: morning, noon, afternoon, evening
  - Sizes: small, medium, large, extra large

ANALYSIS APPROACHES ON GENERIC SERIES

- Dependent on the type of data, only certain analysis methods can be applied
- Methods with arithmetic operations can only be applied on numerical data
- When using comparing methods, nominal data with defined order (e.g. small < medium < large) can be used

NOISY DATA

- Data are often biased with noise (fluctuation by empirial surveys, or from physical detectors)
- Noisy data can only be compared with each other to a limited extent
- One method for improving comparability is to smooth the data

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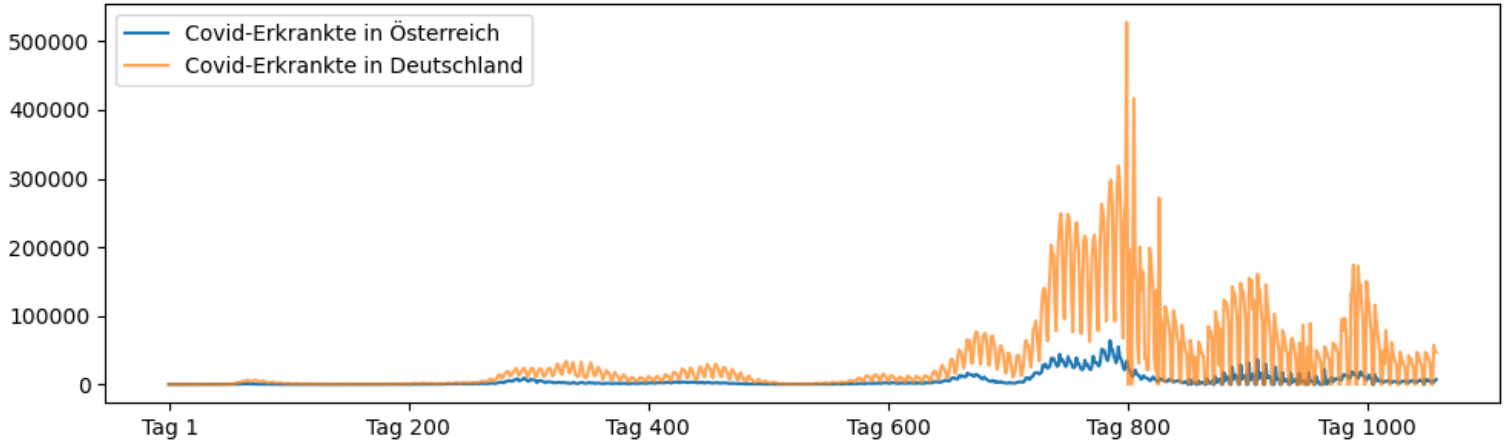
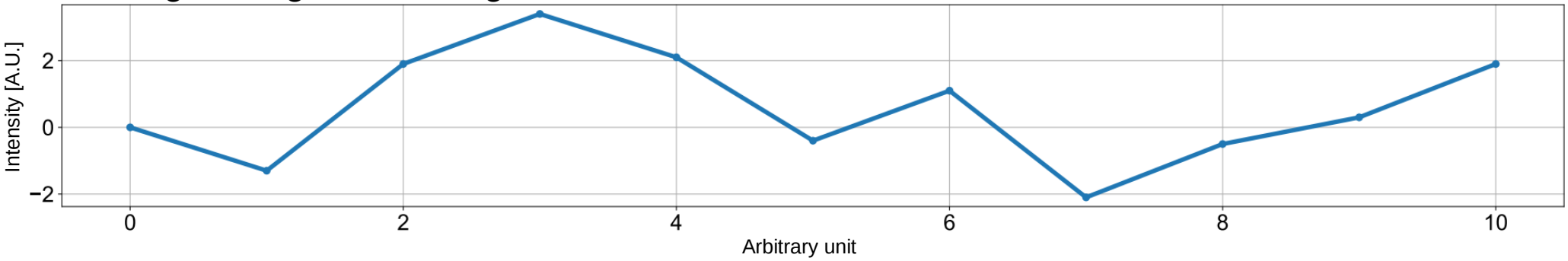


DATA SMOOTHING

- Data smoothing is often a local adjustment of data points dependent on their local neighboring data
- The easiest method for smoothing is the „moving average smoothing“
- When considering a data point, the average of all data points withing a defined window will be assinged to the current data point

DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

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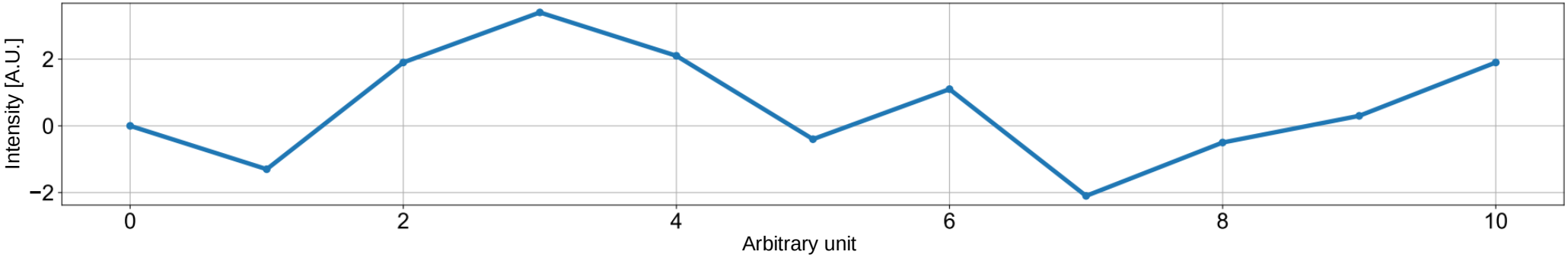
DATA SMOOTHING

moving average smoothing



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DATA SMOOTHING

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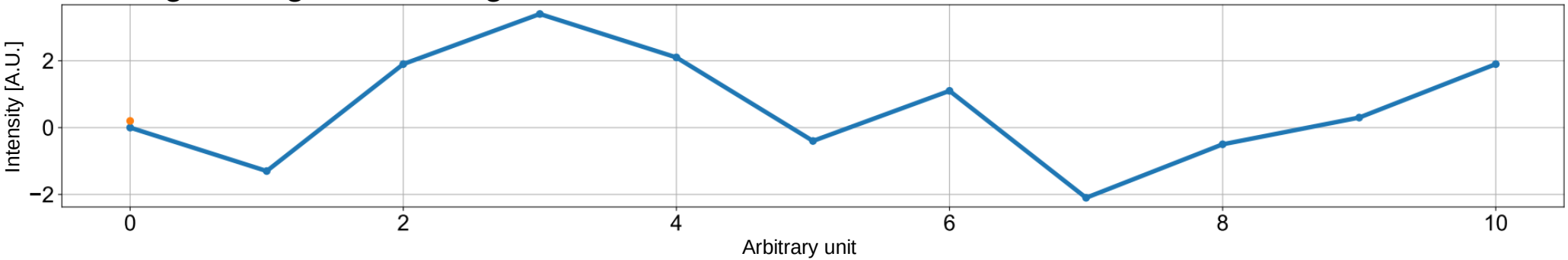

y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2										

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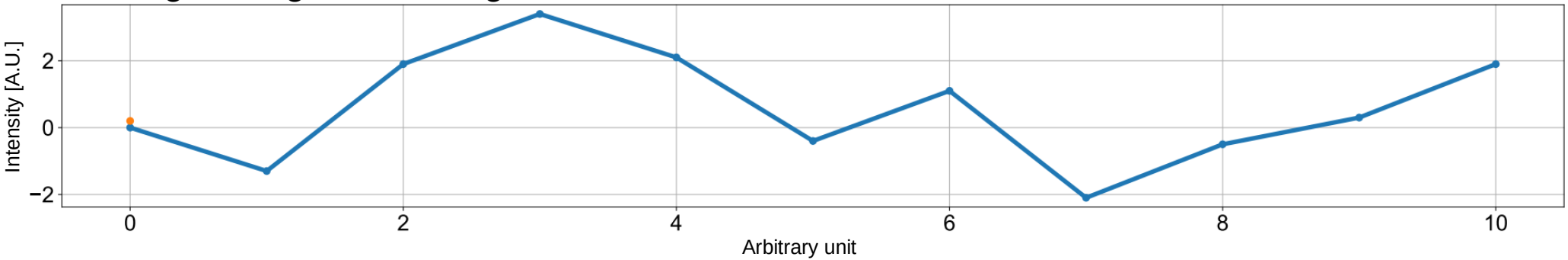
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2										

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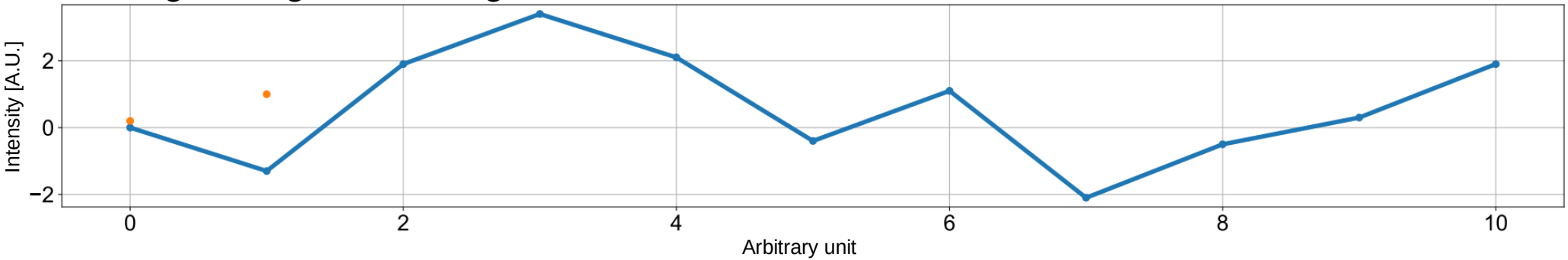
moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0									

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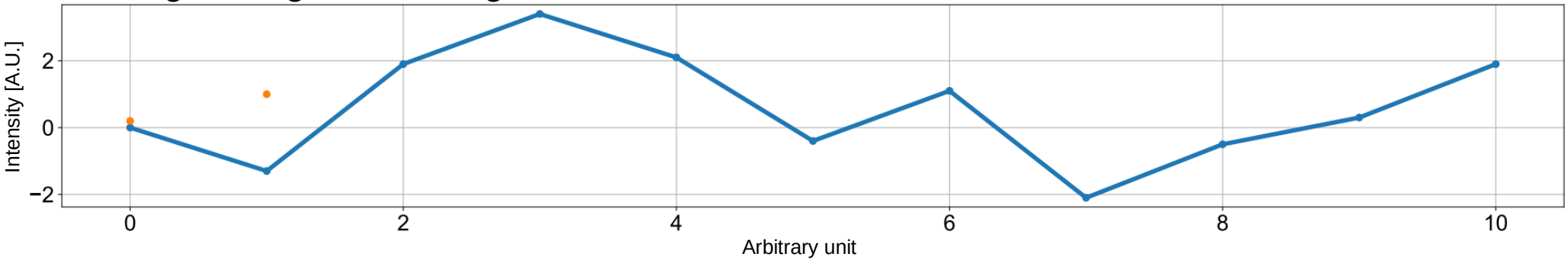
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0									

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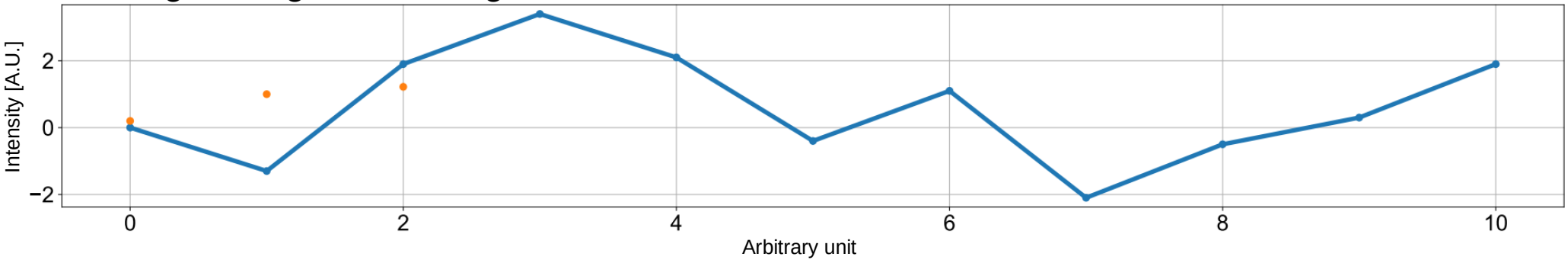
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0	1.22								

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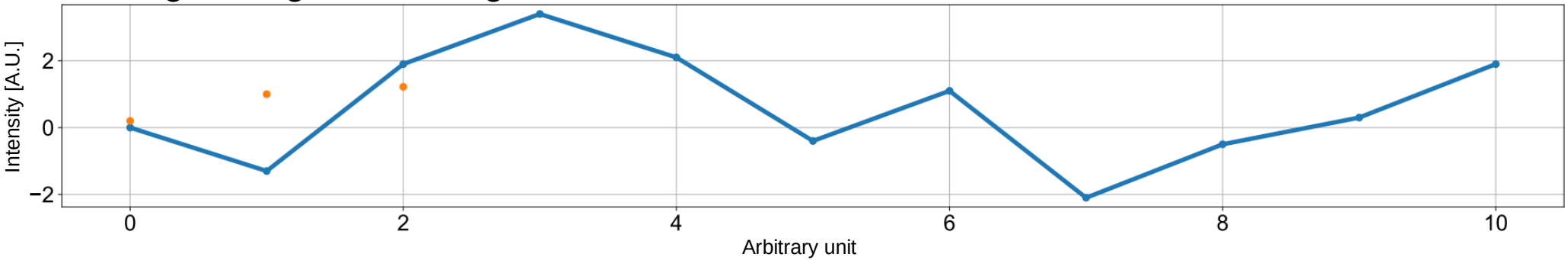
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0	1.22								

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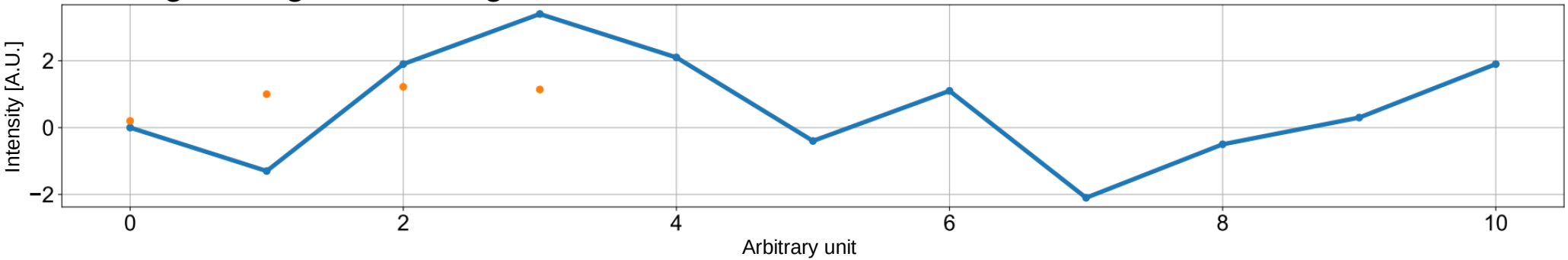
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0	1.22	1.14							

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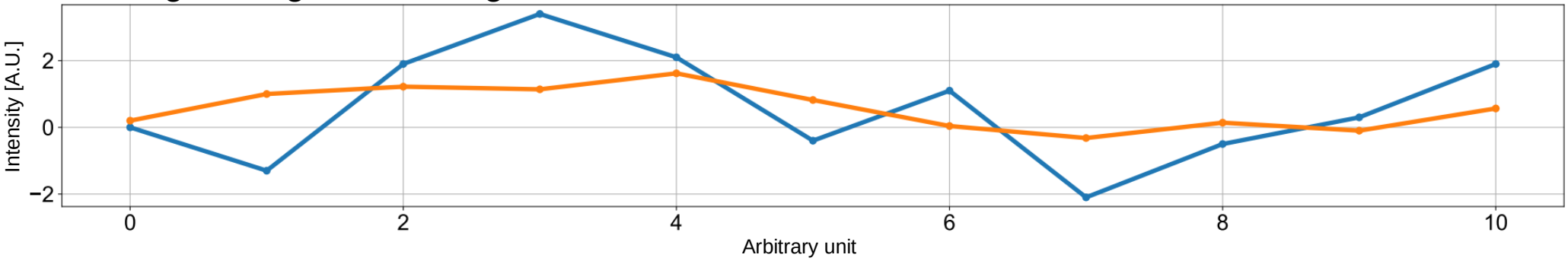
DATA SMOOTHING

moving average smoothing



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

Ø

y’	0.2	1.0	1.22	1.14	1.62	0.82	0.04	-0.31	0.13	-0.1	0.56

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DATA SMOOTHING

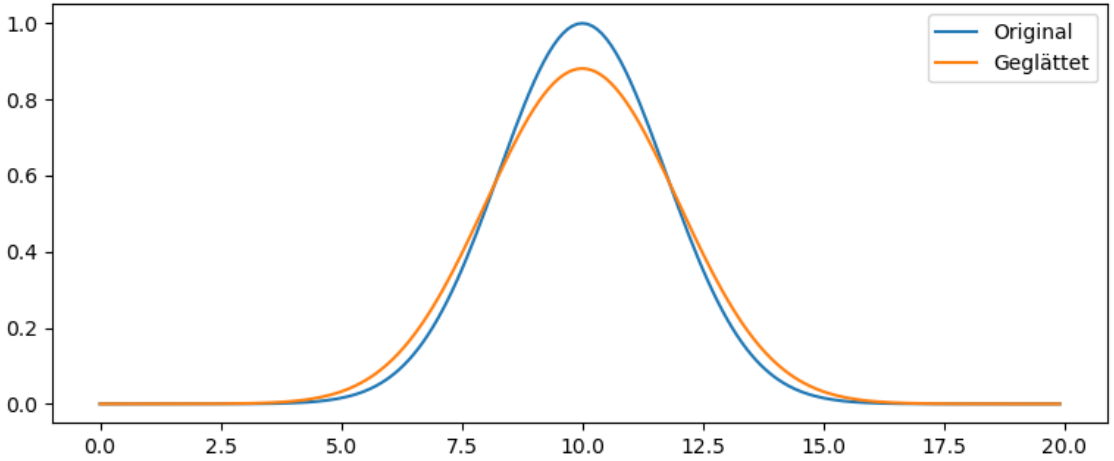
```
def moving_average_smoothing(x, y, window_width):
    y_new, st, en, n, hw = [], 0, 0, len(x), window_width / 2
    for i in range(n):
        while st < n - 1 and x[i] - x[st] > hw: st += 1
        while en < n - 1 and x[en] - x[i] <= hw: en += 1
        y_new.append(sum(y[st : en]) / (en - st + 1))
    return y_new
```

MOVING AVERAGE SMOOTHING

- Simple
- Good when data is very noisy
- Window size must be determined individually

MOVING AVERAGE SMOOTHING

- Simple
- Good when data is very noisy
- Window size must be determined individually
- Not suitable for little noisy data such as chromatograms or MS-spectra
- Peaks are suppressed



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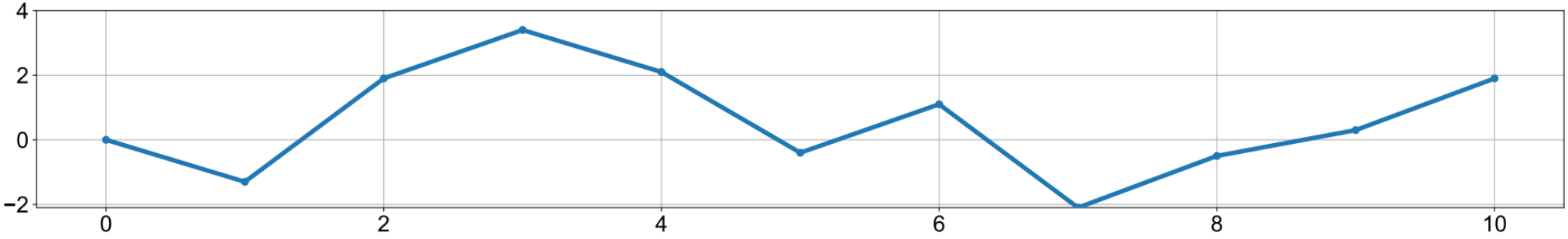
SAVITZKY-GOLAY FILTER

- Smoothing filter from signal processing
- Tries to retain peak tip shape
- Suitable for little noisy data
- Idea: polynomial regression of data points in the local windows

SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

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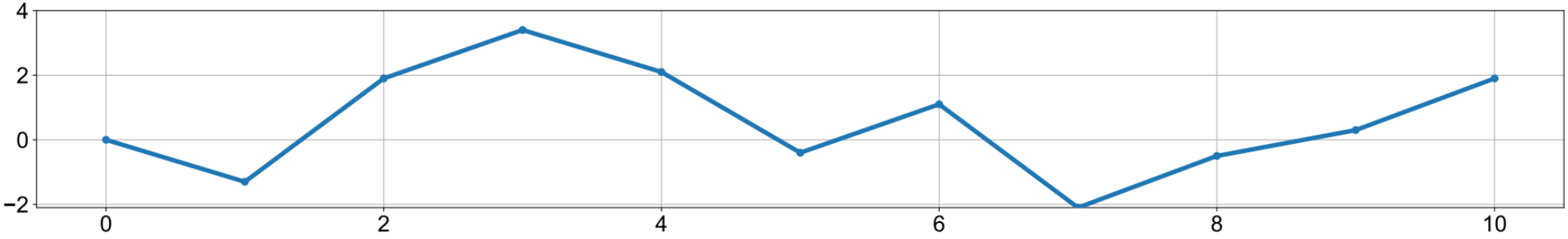
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

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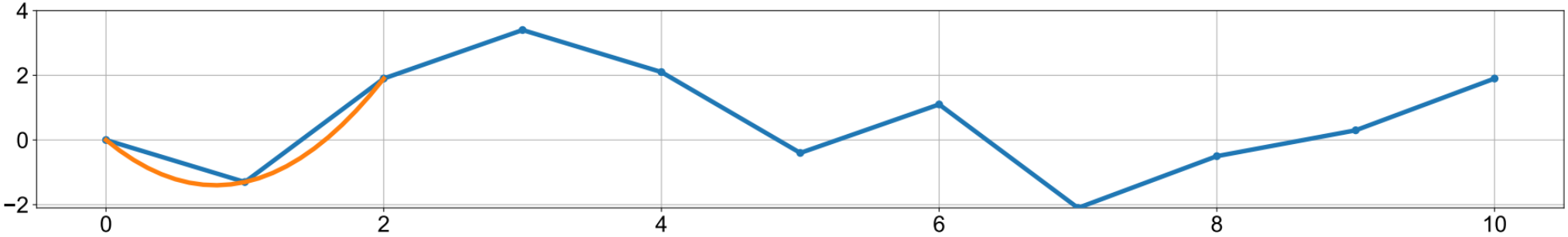

y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x) = 2.25\,x^2 - 3.55\,x$$

y’											

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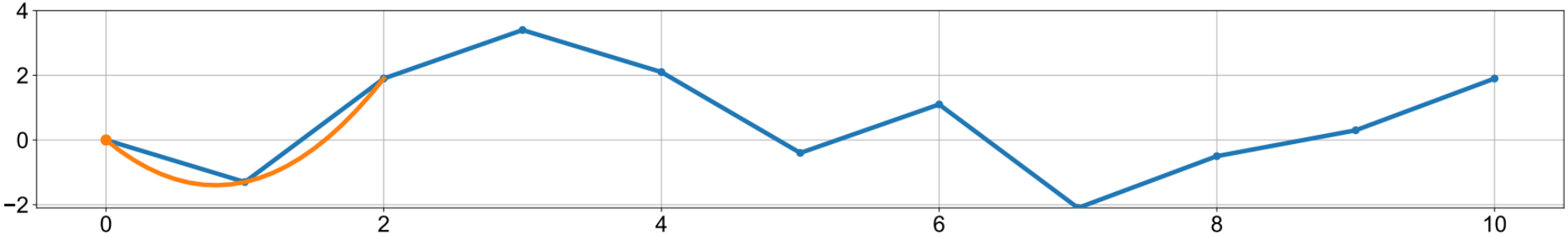
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x)=2.25x^2-3.55x \qquad x=0 \Rightarrow f(x)=0$$

y’	0										

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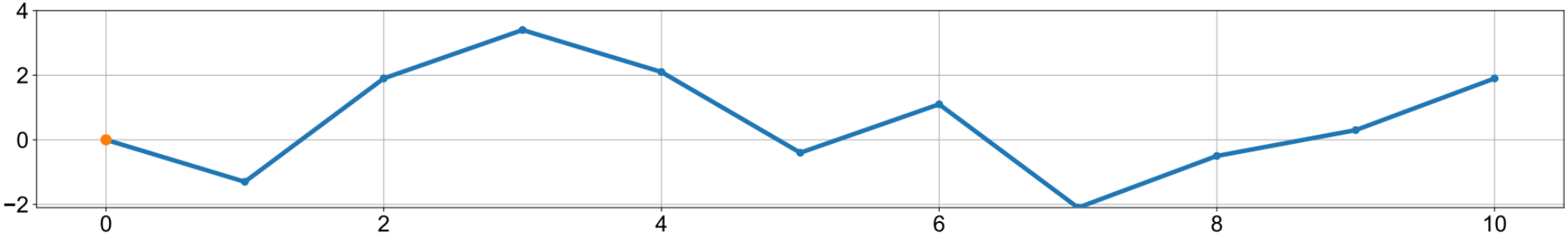
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

y’	0										

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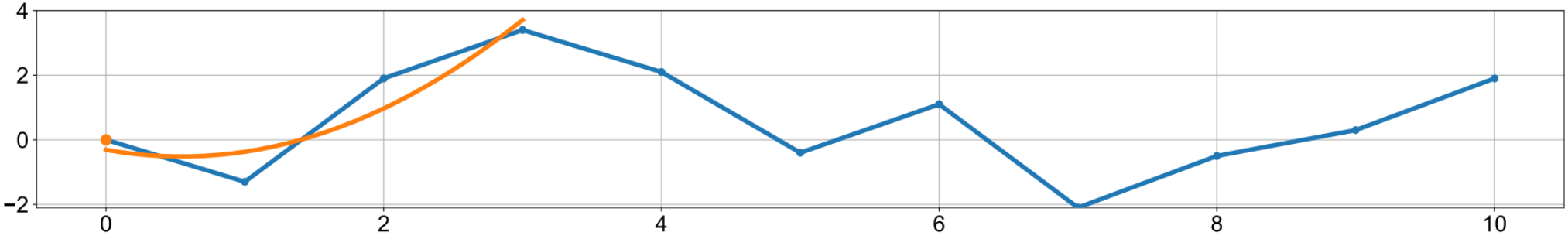

y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x) = 0.7\,x^2 - 0.76\,x - 0.31$$

y’	0										

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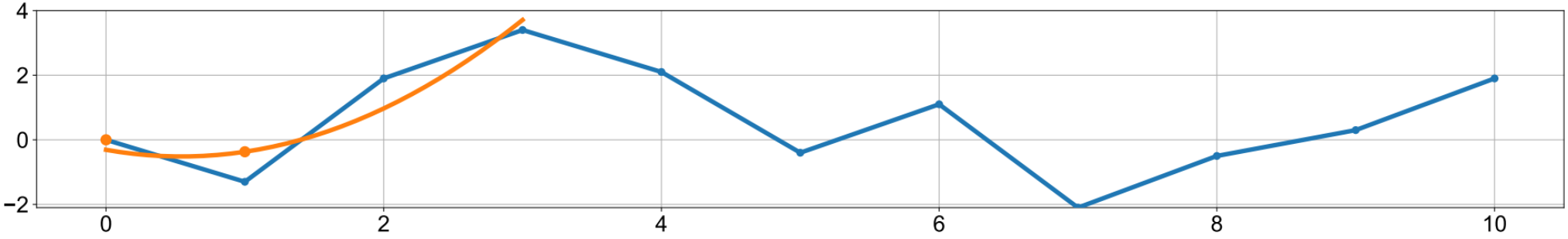
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x)=0.7\,x^2-0.76\,x-0.31 \qquad x=1 \;\Rightarrow\; f(x)=-0.37$$

y’	0	-0.37									

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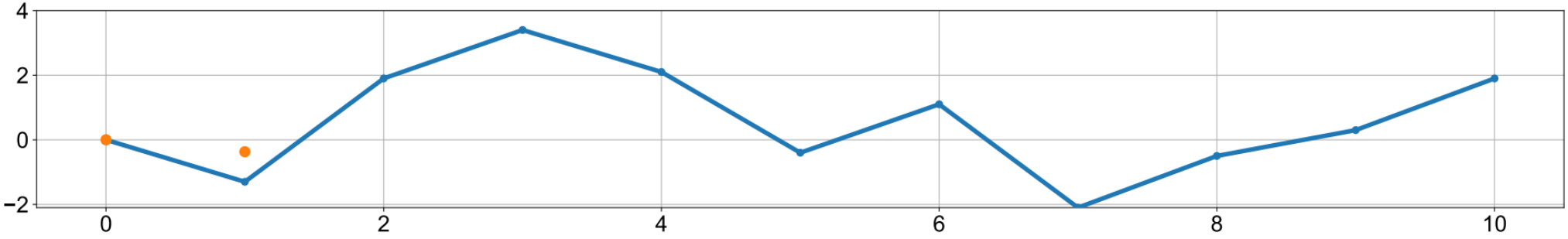
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

y’	0	-0.37									

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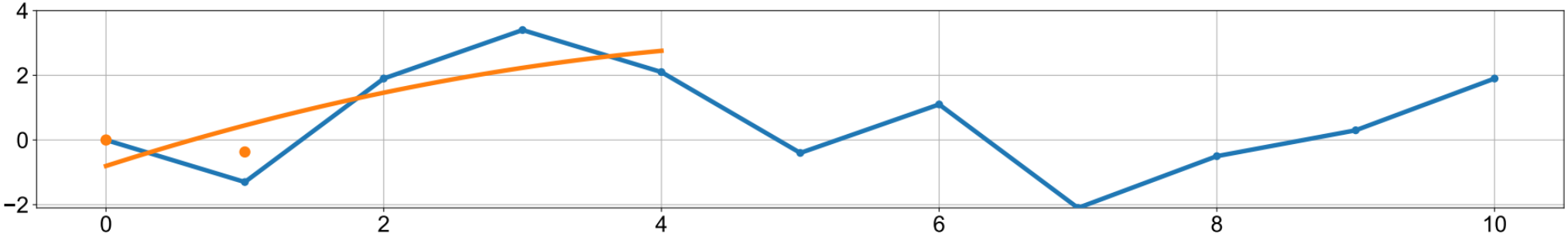
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x) = -0.12\,x^2 + 1.37\,x - 0.80$$

y’	0	-0.37									

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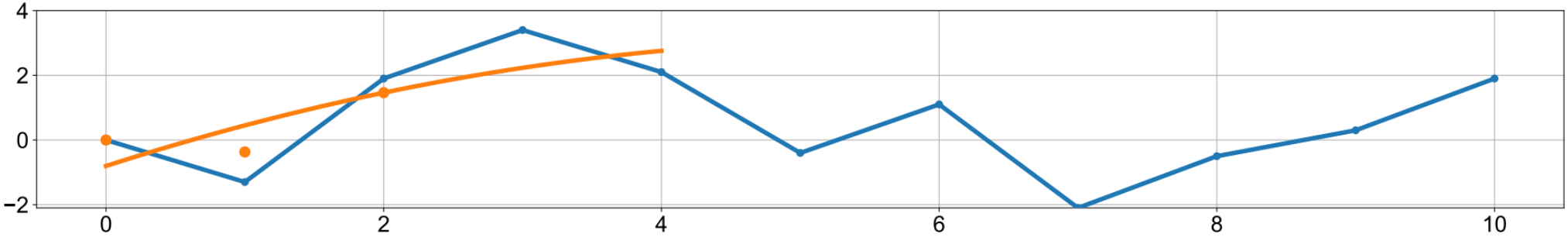
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

$$f(x) = -0.12\,x^2 + 1.37\,x - 0.80 \qquad x = 2 \Rightarrow f(x) = 1.46$$

y’	0	-0.37	1.46								

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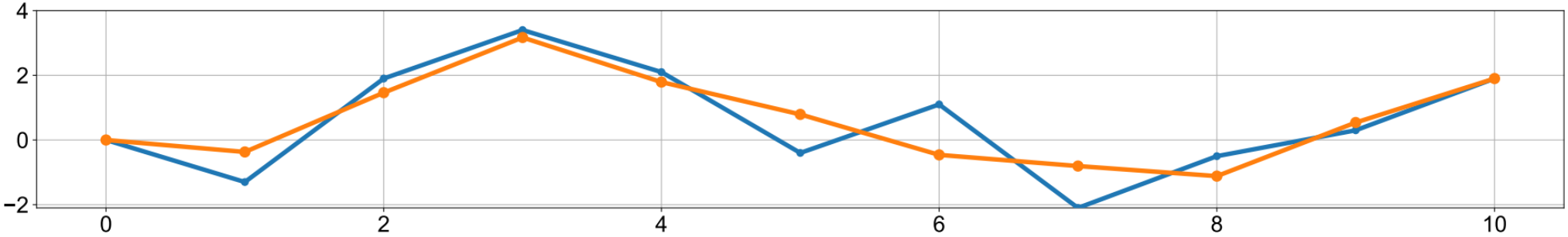
SAVITZKY-GOLAY FILTER



y	0	-1.3	1.9	3.4	2.1	-0.4	1.1	-2.1	-0.5	0.3	1.9

y’	0	-0.37	1.46	3.16	1.79	0.79	-0.45	-0.8	-1.11	0.53	1.89

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DATA SMOOTHING WITH THE SAVITZKY-GOLAY FILTER

```
def sav_gol_filter(x, y, window_width):
    y_new, st, en, n, hw = [], 0, 0, len(x), window_width / 2
    for i in range(n):
        while st < n - 1 and x[i] - x[st] > hw: st += 1
        while en < n - 1 and x[en] - x[i] <= hw: en += 1
        param = np.polyfit(x[st : en], y[st : en], 2)
        y_new.append(param[0] * x[i]**2 + param[1] * x[i] + param[2])
    return y_new
```

COMPARISON OF SERIES

- Remainder: the hierarchical clustering is a distance-based approach
- To cluster our different series, we need to compute pairwise distances
- Requirement: the series need to have the same number of dimensions
  - A measured point at 5 min from the first time series should NOT be compared with a measured point at 7 min from the second time series

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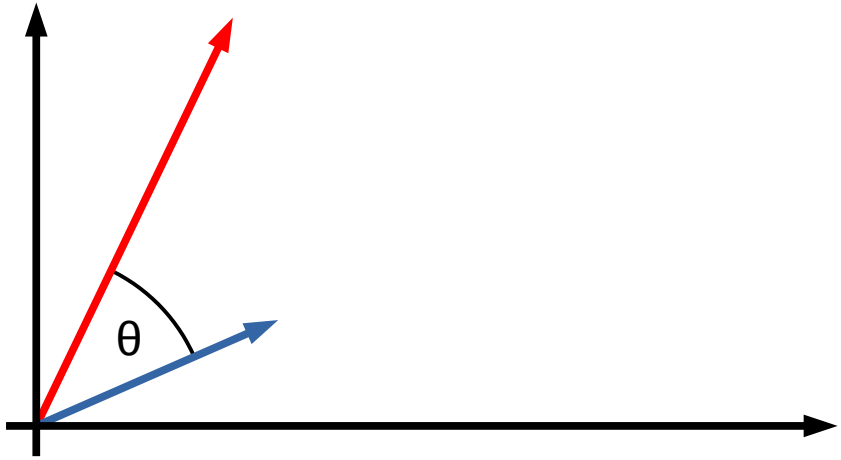
- Remainder: the hierarchical clustering is a distance-based approach
- To cluster our different series, we need to compute pairwise distances
- Requirement: the series need to have the same number of dimensions
  - A measured point at 5 min from the first time series should NOT be compared with a measured point at 7 min from the second time series
- Idea: treat each dimension independently and use Manhattan oder Euclidean distance
- Other solution: Cosine similarity oder Pearson correlation

COSINE SIMILARITY

- Measure for similarity of two vectors from math
- An angle between both *n*-dimensional vectors is computed
- Length of the vectors is irrelevant



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COSINE SIMILARITY

- Measure can have these values:
  - -1 (opposing vector directions)
  - 0 (orthogonal, no similarity)
  - 1 (parallel vectors, full similarity)
  - Everything in between -1 and 1

COSINE SIMILARITY

- Measure can have these values:
  - -1 (opposing vector directions)
  - 0 (orthogonal, no similarity)
  - 1 (parallel vectors, full similarity)
  - Everything in between -1 and 1
  - When the inpute values are all pure positive(or all negative), the measure reaches values between 0 and 1

$$\cos(\theta) = \frac{a \cdot b}{\|a\| \, \|b\|} = \frac{\sum_{i=1}^{n} a_i \cdot b_i}{\sqrt{\sum_{i=1}^{n} a_i^2} \sqrt{\sum_{i=1}^{n} b_i^2}}$$

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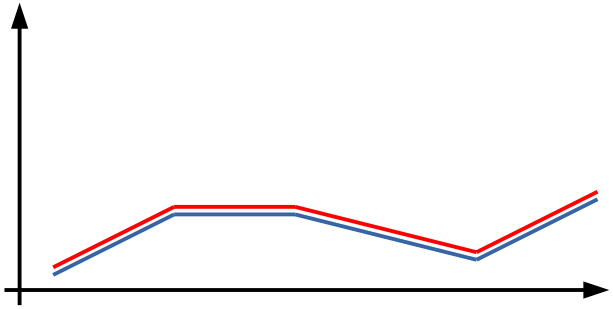
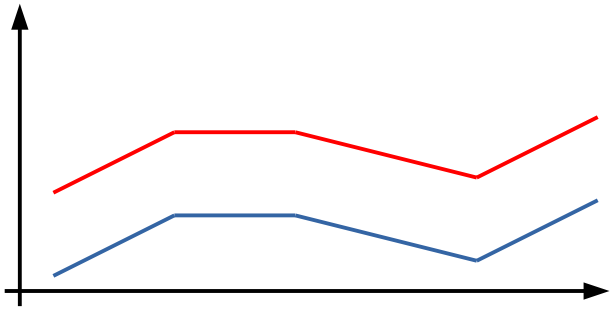
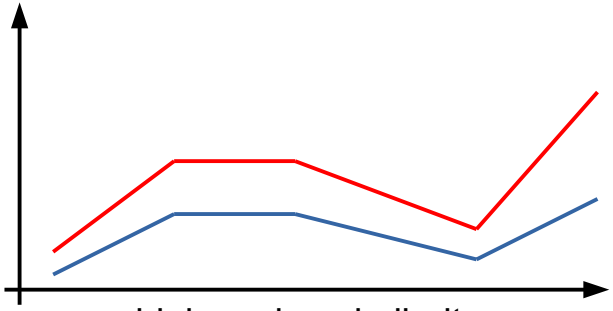
COSINE SIMILARITY, EXAMPLES



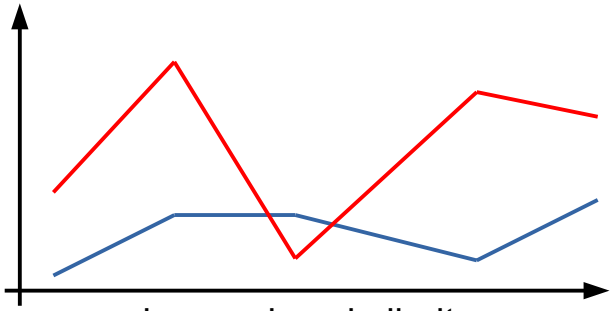
high cosine similarity



low cosine similarity



high cosine similarity



low cosine similarity

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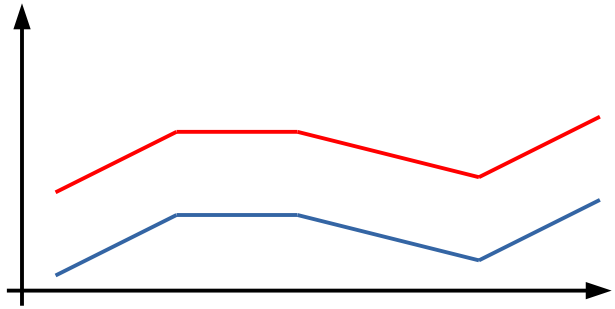
PEARSON CORRELATION

- Two series, which are identical but shifted by a constant value in the y axis have a low cosine similarity



low cosine similarity

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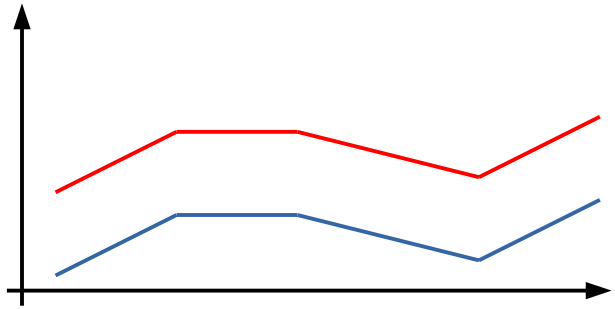
PEARSON CORRELATION

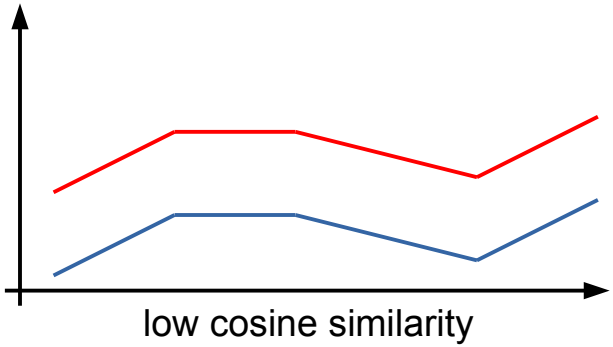
- Two series, which are identical but shifted by a constant value in the y axis have a low cosine similarity
- The Pearson correlation treats this case differently
- correlationen are different to similarities



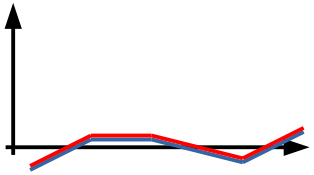
low cosine similarity

PEARSON CORRELATION

- Two series, which are identical but shifted by a constant value in the y axis have a low cosine similarity



- The Pearson correlation treats this case differently
- correlationen are different to similarities
- Both series are shifted at the beginning to have mean value = 0



- Afterwards regular cosine similarity is performed

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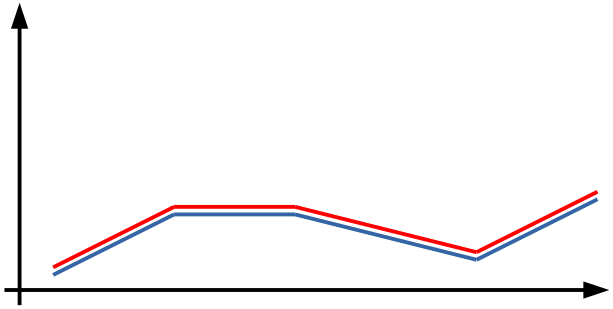
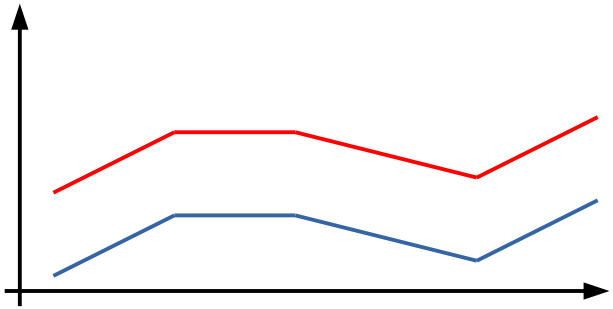
PEARSON CORRELATION, EXAMPLES



high Pearson correlation



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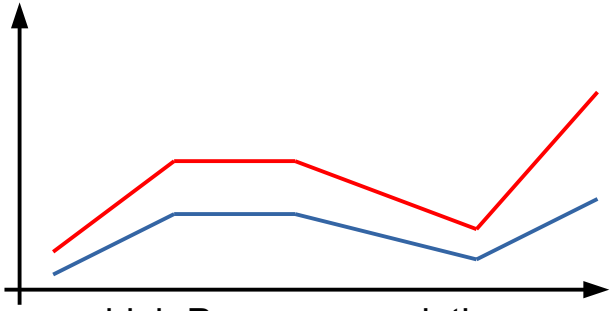

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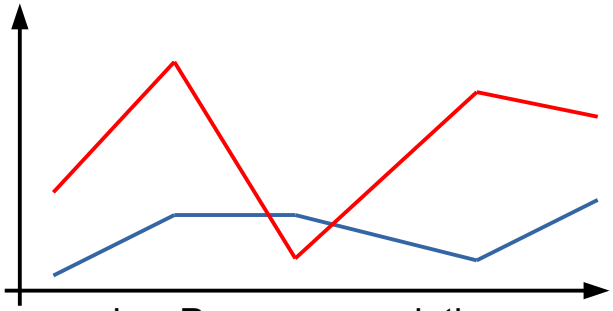

low Pearson correlation

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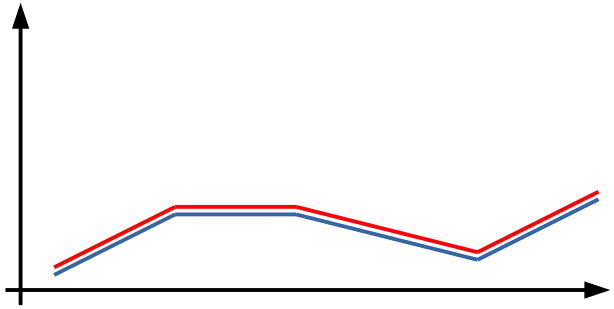
PEARSON CORRELATION, EXAMPLES



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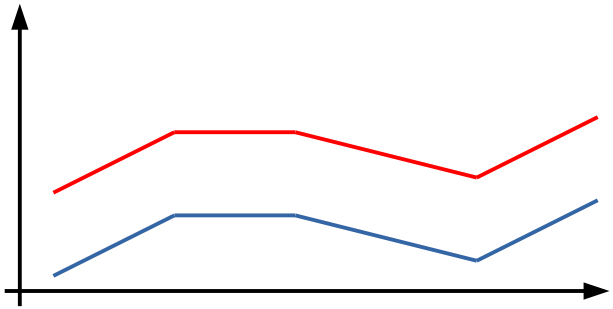

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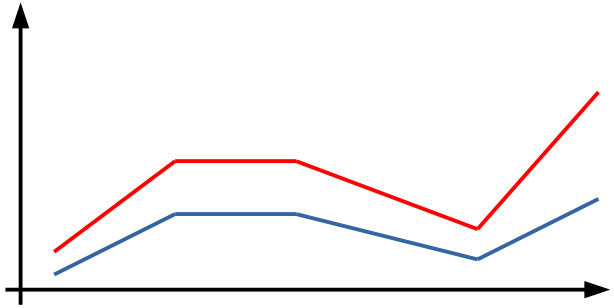

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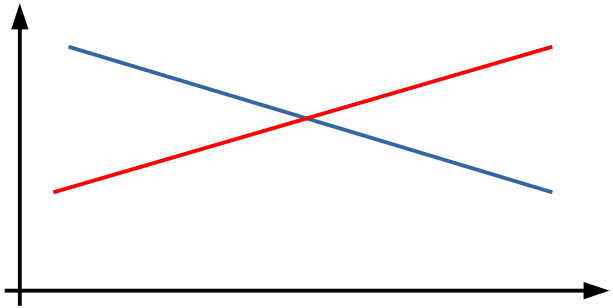

negative Pearson correlation

Although we have only positive values, the Pearson correlationen can reach negative values!

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COMPREHENSIVE ANALYSIS OF ALL TIME SERIES

- Let‘s assume that our table has *n* time series (each series is in a row)
- To compare these series, we need to smooth them upfront
- Afterwards, we need to compute a pairwise distance matrix and visualize it

COMPREHENSIVE ANALYSIS OF ALL TIME SERIES

```
from scipy.spatial.distance import pdist, squareform

covid = pd.read_excel("covid-data-clean.xlsx", index_col = "location")
for i, row in covid.iterrows():
    covid.loc[i] = moving_average_smoothing(range(len(row)), row, 7)

D = 1 - squareform(pdist(covid, "cosine"))   # create a pairwise dist. matrix

cb = plt.imshow(D)
plt.colorbar(cb)
plt.show()
```

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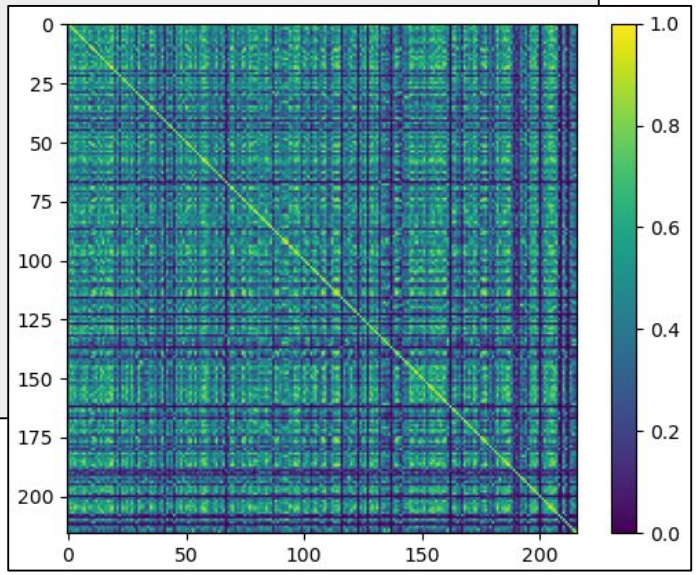

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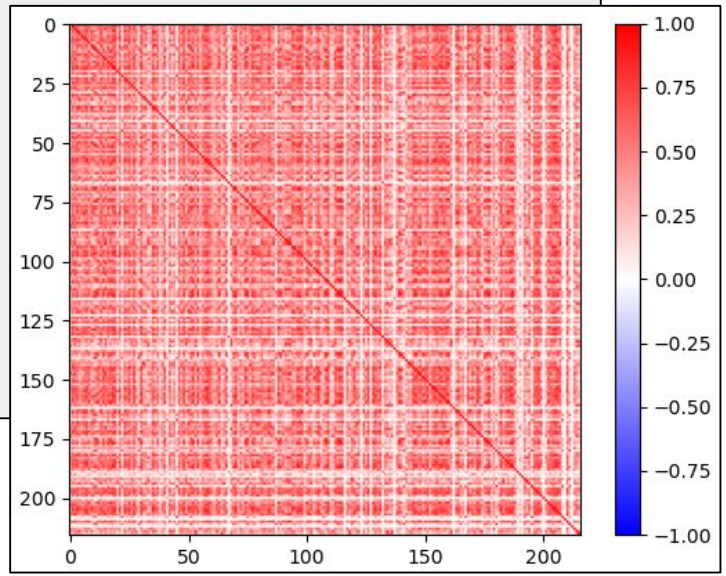

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COMPREHENSIVE ANALYSIS OF ALL TIME SERIES

- The computed distance matrix looks very weird and we cannot recognize here anything
- The rows and columns need to be sorted that similar rows are being put together

COMPREHENSIVE ANALYSIS OF ALL TIME SERIES

- The computed distance matrix looks very weird and we cannot recognize here anything
- The rows and columns need to be sorted that similar rows are being put together
- The sorting can be again performed with hierarichical clustering, as discussed in the previous lecture

COMPREHENSIVE ANALYSIS OF ALL TIME SERIES

```
covid = pd.read_excel("covid-data-clean.xlsx", index_col = "location")
for i, row in covid.iterrows():
    covid.loc[i] = moving_average_smoothing(range(len(row)), row, 7)
# create a pairwise dist. matrix for visualization
D = 1 - squareform(pdist(covid, "cosine"))

# create a clustering and sort the observations accordingly
lkg = hierarchy.linkage(covid, method = 'average', metric = "cosine")
labels_sorting = hierarchy.leaves_list(lkg)

D = D[labels_sorting]      # rowwise sorting
D = D[:, labels_sorting]   # columnwise Sorting
cb = plt.imshow(D, cmap = "bwr", vmin = -1, vmax = 1)
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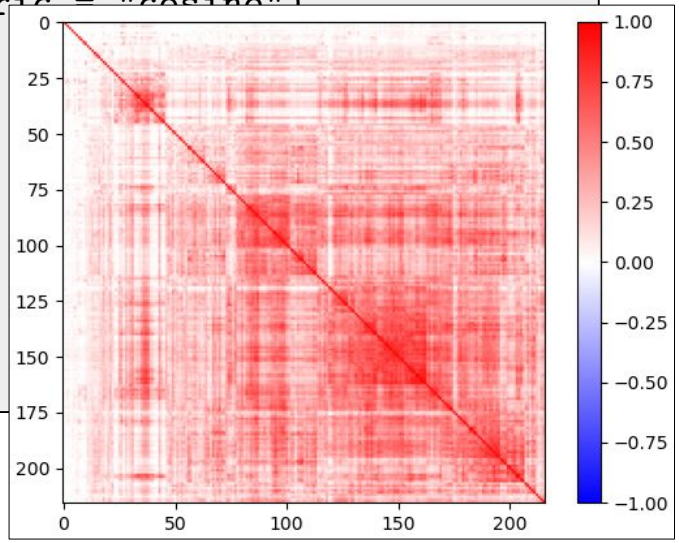

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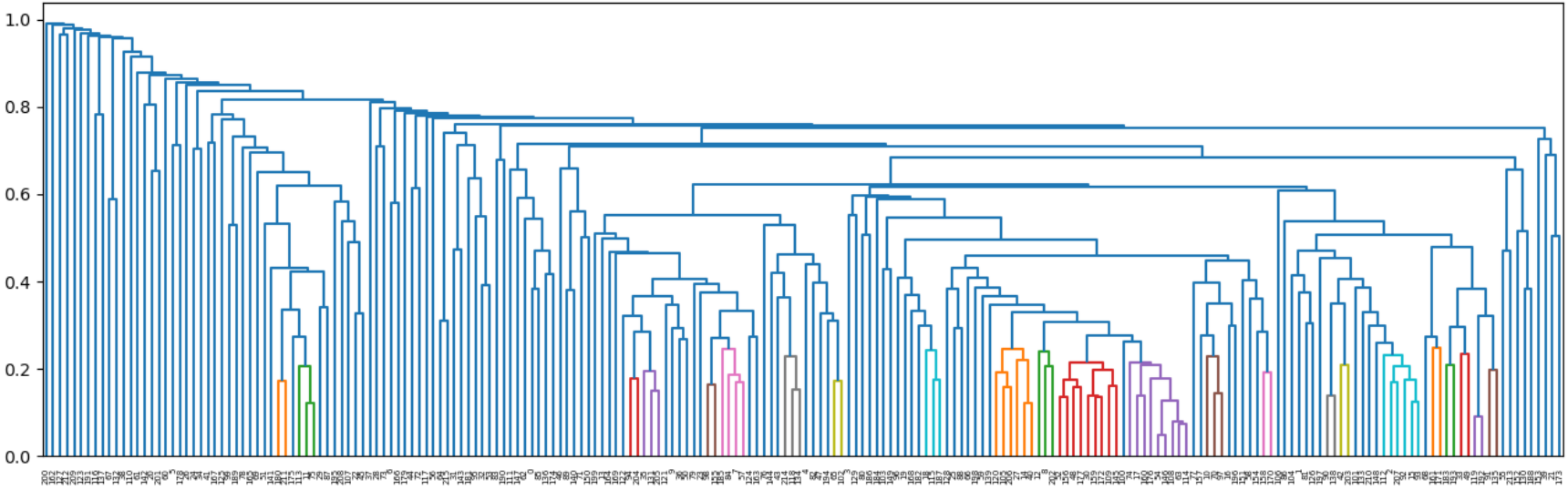
GROUPING OF TIME SERIES

- The hierarchical clustering can be used, to group the time series
- We want to have groups of time series that each series has at least a similarity of x % to all others within a group
- Here, we can apply the knowledge we got from the previous lecture, let‘s have a look again

GROUPING OF TIME SERIES

```
lkg = hierarchy.linkage(covid, method = 'average', metric = "cosine")
# pairwise similarity of at least 75% is 1 - 0.75 = 0.25 distance
dn = hierarchy.dendrogram(lkg, color_threshold = 0.25)
plt.show()
```



GROUPING OF TIME SERIES

```
# create a cluster list
clusters = hierarchy.fcluster(lkg, t = 0.25, criterion = 'distance')
print(clusters)
# [50 123 127  88  86  10  36  70 103 ... 28 161 129  18   2 142  79  41]

# create a dictionary: group index → list of observation row indexes
groups = {c: [] for c in set(clusters)}
for i, c in enumerate(clusters): groups[c].append(i)

# print groups in the console
for k, v in groups.items():
    print(k, [covid.index[i] for i in v])
```

GROUPING OF TIME SERIES

```
...
19 ['Austria', 'Germany', 'Liechtenstein']
70 ['Argentina', 'Dominican Republic', 'Guyana', 'Suriname']]
102 ['Bahrain', 'Brazil', 'Chile', 'Kuwait', 'Maldives', 'Uruguay']
103 ['Armenia', 'Azerbaijan', 'Ukraine']
104 ['Bulgaria', 'Croatia', 'Czechia', 'Lebanon', 'North Macedonia', 'Poland',
'Romania', 'Serbia', 'Slovenia']
105 ['Belarus', 'Denmark', 'Estonia', 'Georgia', 'Latvia', 'Lithuania', 'Norway',
'Russia', 'Slovakia']]
127 ['Algeria', 'Bangladesh', 'Iran', 'Iraq', 'Libya', 'Uzbekistan']
...
```

THE BIG PICTURE OF THE TIME SERIES ANALYSIS

- We have different components, that we can visualize:
  - Heatmap
  - Hierarchical clustering
  - Grouping
- It would be awesome, when everything could be united in one figure
- The function `clustermap` from the `seaborn` library can be used for that

THE BIG PICTURE OF THE TIME SERIES ANALYSIS

```
import seaborn as sns      # import Seaborn library

covid = pd.read_excel("covid-data-clean.xlsx", index_col = "location")
for i, row in covid.iterrows():
    covid.loc[i] = moving_average_smoothing(range(len(row)), row, 7)

D = 1 - squareform(pdist(covid, "cosine"))   # create pairwise dist. matrix

cm = pd.DataFrame(D, index = covid.index, columns = covid.index)
sns.clustermap(cm, method = "average", metric = "cosine", cmap = "bwr",
vmin = -1, vmax = 1)
plt.show()
```

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plt.show()
```

Different tasks which use the same metric but don‘t need to

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THE BIG PICTURE OF THE TIME SERIES ANALYSIS

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import seaborn as sns      # import Seaborn

covid = pd.read_excel("covid-data-clean.xls
for i, row in covid.iterrows():
    covid.loc[i] = moving_average_smoothing

D = 1 - squareform(pdist(covid, "cosine"))

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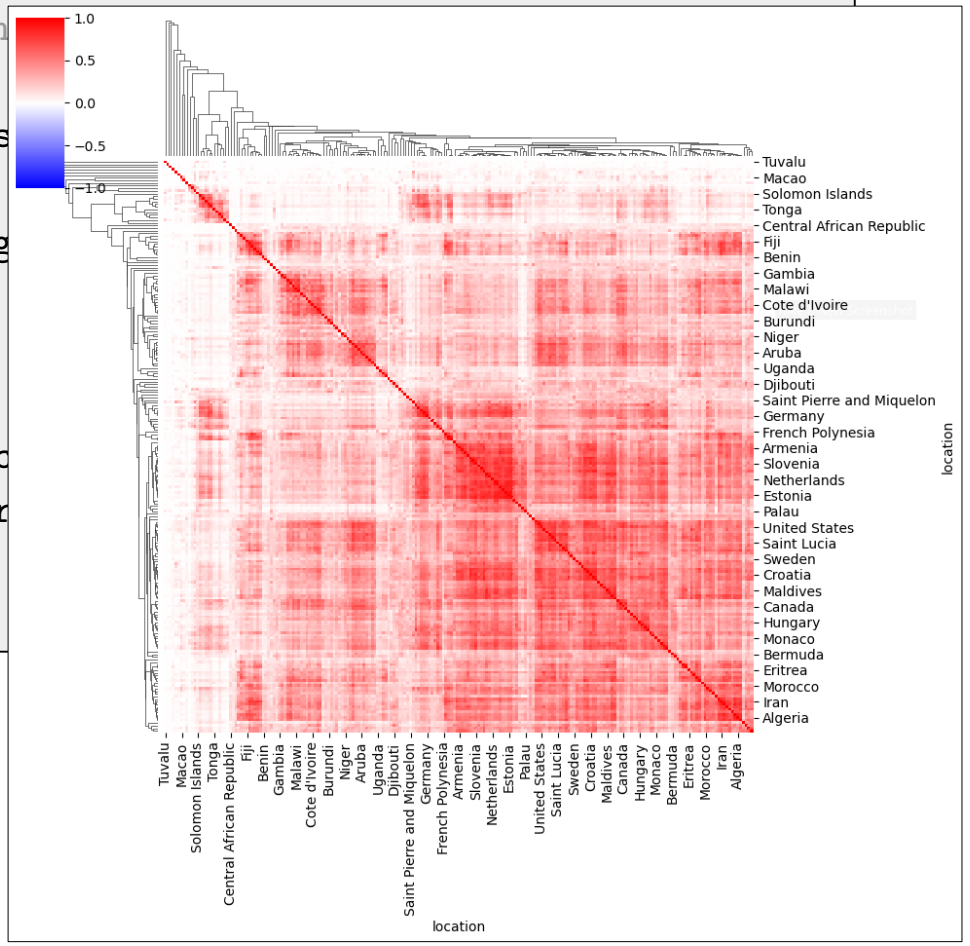

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THE BIG PICTURE OF THE TIME SERIES ANALYSIS

- The grouping has to be added yet
- But it is dependent on the sorting of the clustering

THE BIG PICTURE OF THE TIME SERIES ANALYSIS

- The grouping has to be added yet
- But it is dependent on the sorting of the clustering
- We applied two different clusterings from different libraries:
  - with `hierarchy.linkage`
  - with `sns.clustermap`
- Since we determine the grouping with the result of `hierarchy.linkage`, we insert it again in `sns.clustermap`

THE BIG PICTURE OF THE TIME SERIES ANALYSIS

```
import matplotlib.colors as mcolors

lkg = hierarchy.linkage(covid, method = 'average', metric = "cosine")
clusters = hierarchy.fcluster(lkg, t = 0.25, criterion = 'distance')

# create a list of colors independent of the time series groups
colors = [v for k, v in mcolors.TABLEAU_COLORS.items()]
group_colors = [colors[c % len(colors)] for c in clusters]

D = 1 - squareform(pdist(covid, "cosine"))
cm = pd.DataFrame(D, index = covid.index, columns = covid.index)
sns.clustermap(cm, row_linkage = lkg, col_linkage = lkg, cmap = "bwr", vmin = -1,
vmax = 1, row_colors = group_colors)

plt.show()
```

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THE BIG PICTURE OF THE TIME SERIES ANALYSIS

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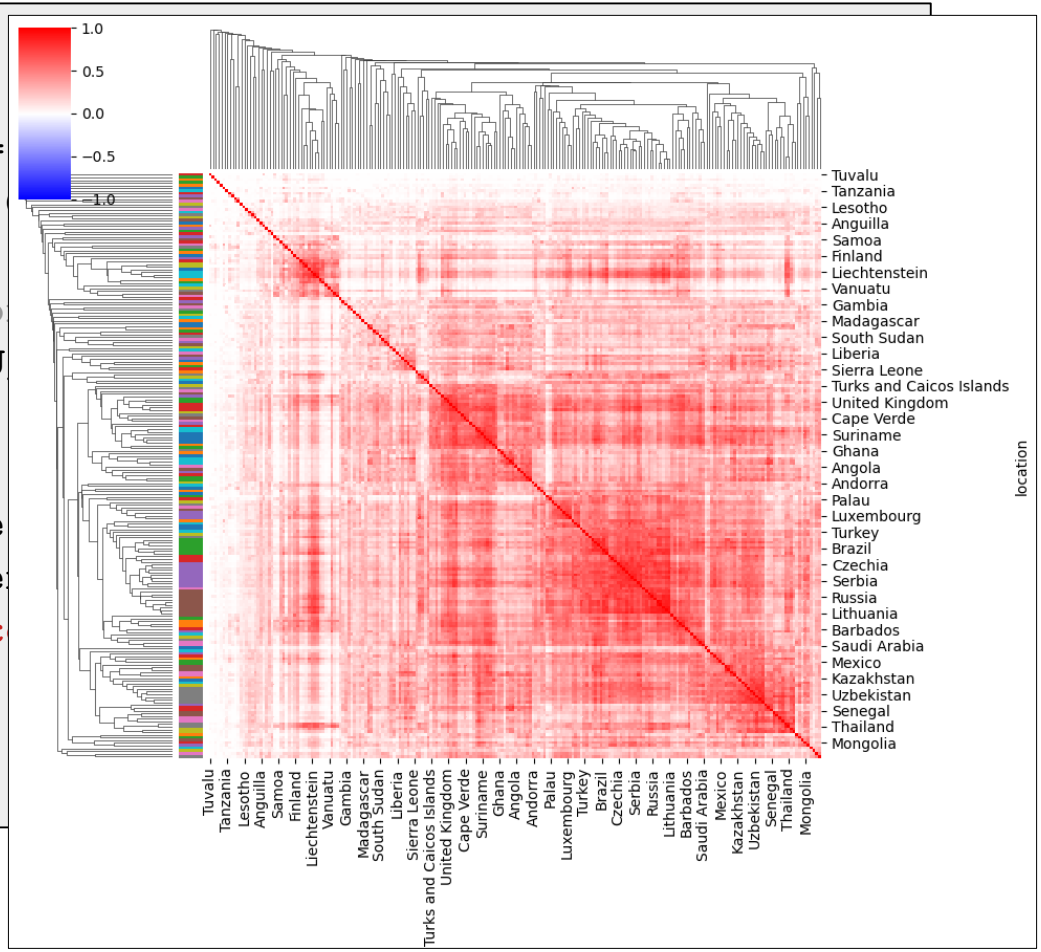

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